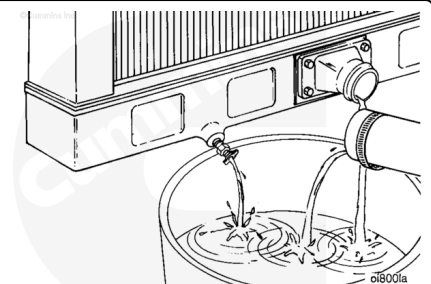


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



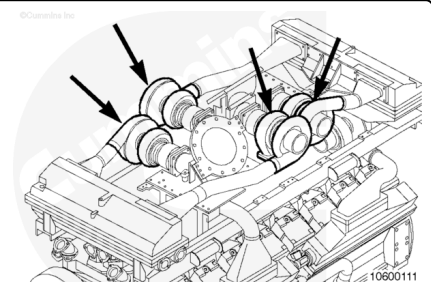
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

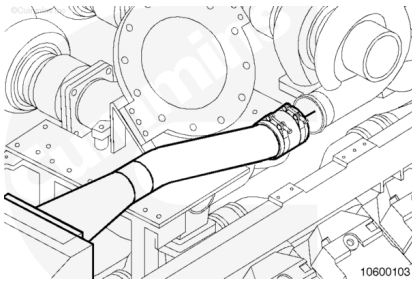
Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

Remove

The low-pressure turbochargers are used **only** on the QSK60 two-stage engine configuration, and are mounted to the two-stage frame assembly above the high-pressure turbochargers.

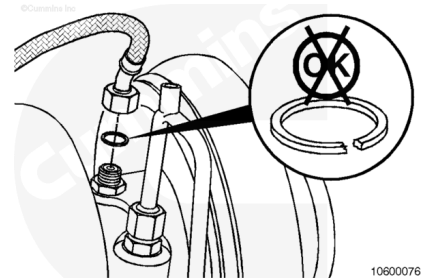


Determine which of the four low-pressure turbochargers requires removal. If it is a left bank rear or right bank front turbocharger, loosen the clamps on the hose between the turbocharger and the intercooler connection.



Disconnect the oil supply hose.

Verify the o-ring is in the correct position and has **not** been damaged.

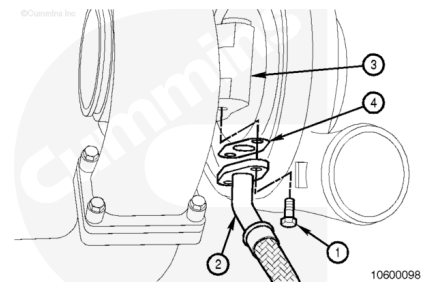


Remove the two oil drain line capscrews (1).

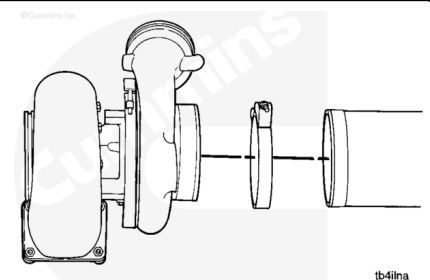
Remove the oil drain line (2) from the turbocharger (3).

Discard the gasket (4).

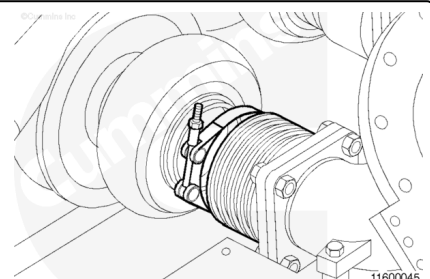
Discard the hose if it is cracked or frayed.



Remove the inlet piping to the turbocharger.

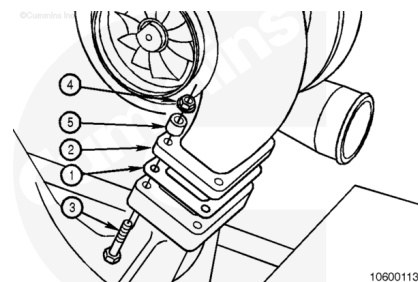


Remove the V-band clamp between the exhaust bellows and the turbocharger.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of capscrews for assembly.

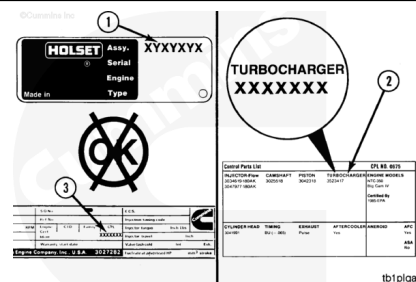
Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

Check for Correct Component

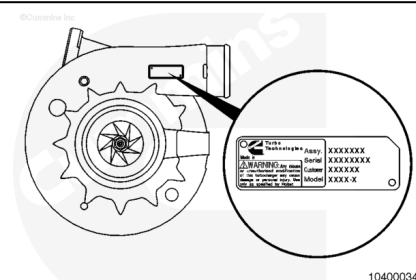
Compare the assembly number (1) on the turbocharger data plate with the turbocharger specified in the Control Parts List (CPL), Bulletin 3379133 (2).

The CPL number for each engine is listed on the engine dataplate (3).



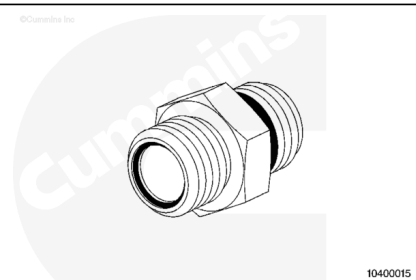
Install

Holset® turbocharger(s) are used on the QSK45 and the QSK60 engines. Refer to the data tag on the turbocharger to determine the model.



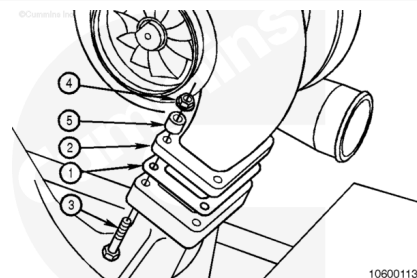
All turbocharger fittings are the flat face o-ring type. Make sure the o-ring is in place before attaching the hose(s) to the fittings.

The Holset® turbocharger uses a 9/16-18 UNF straight thread, o-ring type of oil supply fitting.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows

Apply an anti-seize compound to the mounting capscrew threads.

Install a new turbocharger gasket (1).

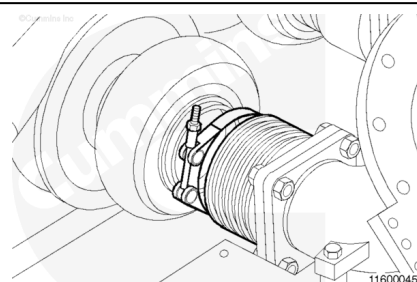
Install the turbocharger (2) with capscrews (3), spacers (5), and flange nuts (4).

Tighten the capscrews.

Torque Value: 55 n•m [40 ft-lb]

Install and tighten the V-band clamp between the exhaust bellows and turbocharger.

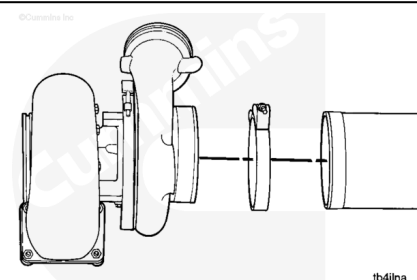
Torque Value: 9 n•m [80 in-lb]



Install the inlet piping and clamp to the turbocharger.

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]



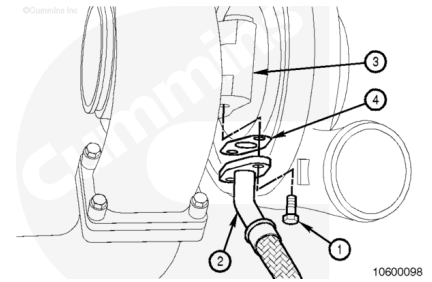
Install a new gasket (4) on the turbocharger oil drain line.

Lubricate the oil drain line mounting capscrews with anti-seize compound, Part Number 3824879.

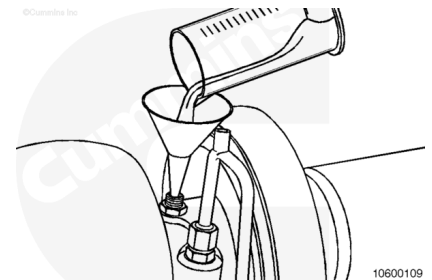
Install the drain line (2) and two drain line mounting capscrews (1) to the turbocharger (1).

Tighten the turbocharger drain line mounting capscrews.

Torque Value: 45 n•m [33 ft-lb]



Pour 50 to 60 cc [2.0 to 3.0 oz] of clean engine oil into the turbocharger oil supply fitting.

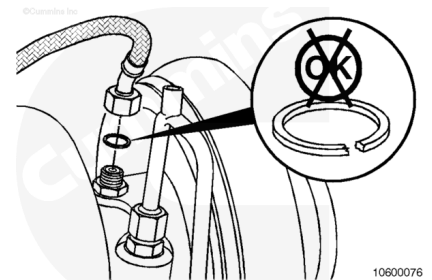


Make sure the o-ring is in the correct position and is **not** damaged.

Install the oil supply hose.

Tighten the hose fitting.

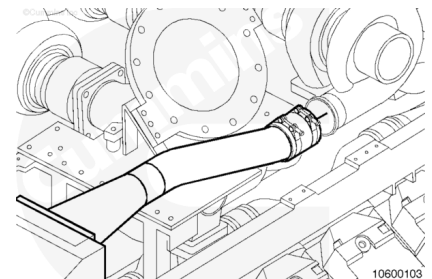
Torque Value: 54 n•m [40 ft-lb]



Install the hose and two clamps between the low-pressure turbocharger outlet and the intercooler intake crossover.

Tighten the clamps.

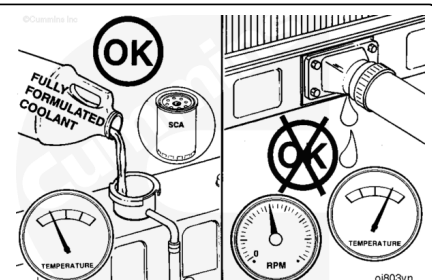
Torque Value: 8 n•m [71 in-lb]



Finishing Steps

Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

Operate the engine to the required coolant temperature. Check for leaks.



Measurements		
	celsius	fahrenheit
Operating Coolant Temperature	71	160